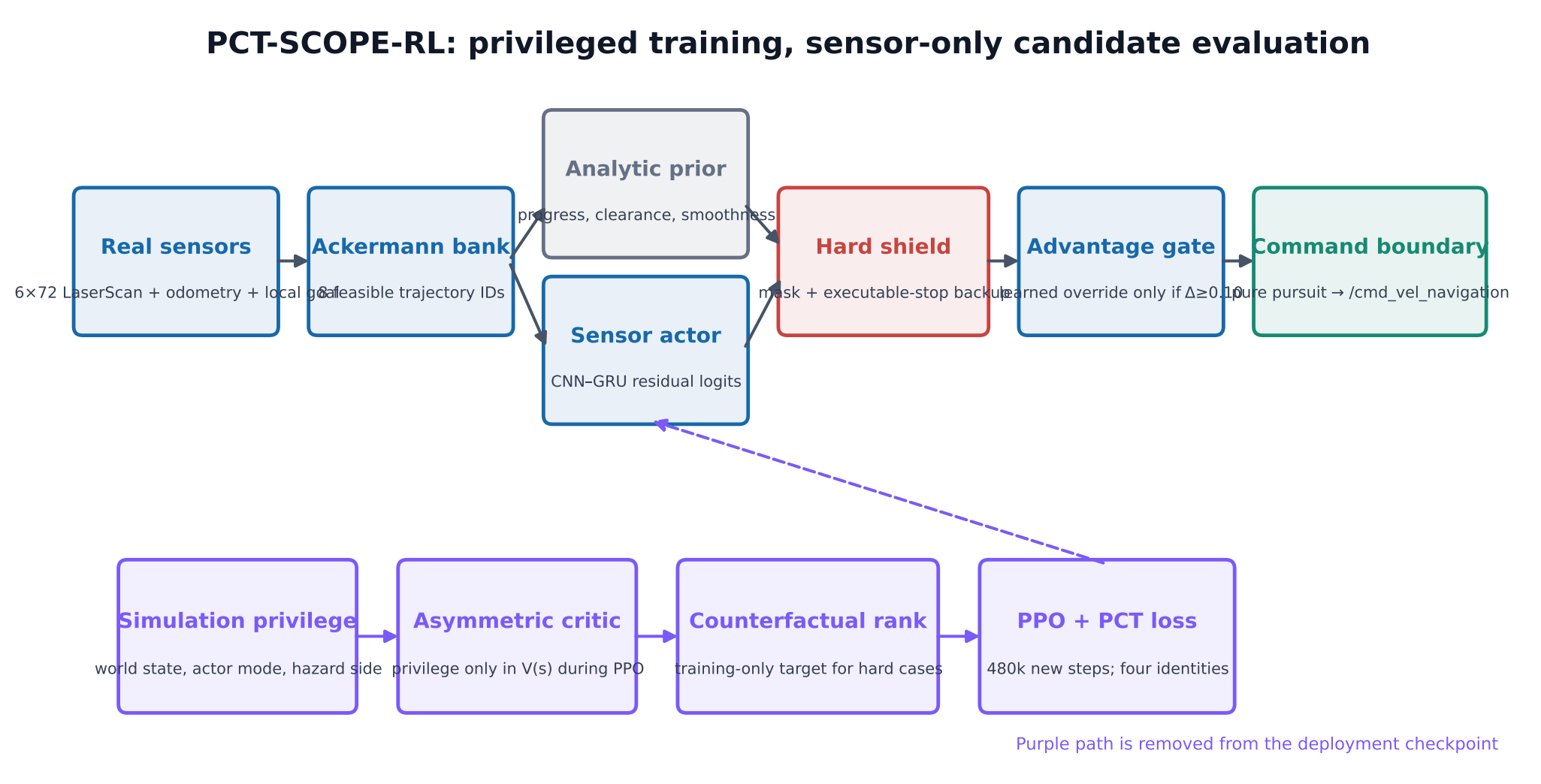
PCT-SCOPE-RL: privileged training, sensor-only candidate evaluation
Real sensors
6×72 LaserScan + odometry + local goal
Ackermann bank
81 feasible trajectory IDs
Analytic prior
progress, clearance, smoothness
Sensor actor
CNN–GRU residual logits
Hard shield
mask + executable-stop backup
Advantage gate
learned override only if Δ≥0.10
Command boundary
pure pursuit → /cmd_vel_navigation
Simulation privilege
world state, actor mode, hazard side
Asymmetric critic
privilege only in V(s) during PPO
Counterfactual rank
training-only target for hard cases
PPO + PCT loss
480k new steps; four identities
Purple path is removed from the deployment checkpoint